Supplemental Results from Another Simulation Study Where the Start Values are Generated Following Hipp & Bauer (2006) Framework

Table 1

Fixed and Manipulated Parameter Values

		Class 1	Class 2
Fixed Conditions	1. Parameters:		
	Mean Intercept (β_{11})	2.00	
	Mean Slope 1 (β_{21})	0.00	
	Mean Slope 2 (β_{41})	1.25	
	Variance of Intercept (φ_{11})	0.40	0.40
	Variance of Slope 1 (φ_{22})	0.20	0.20
	Variance of Slope 2 (φ_{33})	0.20	0.20
	Variance of Knot (φ_{44})	0.20	0.20
	All Covariances among Random effects	0.00	0.00
	2. Number of Time Points (1 to 9)	9.00	9.00
	3. Class Proportion	0.50	0.50
Manipulated Conditions	1. Knot Location		
	Early	3.00	3.25
	Middle	5.00	5.25
	2. Residual Variance		
	Low	0.15	0.15
	High	0.50	0.50
	3. Sample Size		
	N = 200	100.00	100.00
	N = 400	200.00	200.00
	4. Class Separation		
	MD = 1.18		
	Mean Intercept (β_{12})		2.25
	Mean Slope 1 (β_{22})		0.35
	Mean Slope 2 (β_{42})		1.50
	MD = 2.42		
	Mean Intercept (β_{12})		3.10
	Mean Slope 1 (β_{22})		0.50
	Mean Slope 2 (β_{42})		1.75

Table 2

Model Convergence Rate Across all Conditions

Sample Size	Residual Variance	Knot Locations	MD	Convergence Rate
200	0.15	[3,3.25]	1.18	51%
200	0.15	[3,3.25]	2.42	40%
200	0.15	[5,5.25]	1.18	76%
200	0.15	[5,5.25]	2.42	72%
200	0.50	[3,3.25]	1.18	16%
200	0.50	[3,3.25]	2.42	12%
200	0.50	[5,5.25]	1.18	57%
200	0.50	[5,5.25]	2.42	43%
400	0.15	[3,3.25]	1.18	59%
400	0.15	[3,3.25]	2.42	67%
400	0.15	[5,5.25]	1.18	75%
400	0.15	[5,5.25]	2.42	74%
400	0.50	[3,3.25]	1.18	21%
400	0.50	[3,3.25]	2.42	24%
400	0.50	[5,5.25]	1.18	55%
400	0.50	[5,5.25]	2.42	54%

Table 3

Average Parameter Bias Across all Conditions

Parameter	Class 1	Class 2
β_1	0.217	-0.204
β_2	0.101	-0.130
β_4	0.095	-0.084
γ	0.051	0.027
φ_{11}	0.117	-0.005
φ_{22}	0.025	0.155
φ_{33}	0.009	0.021
φ_{44}	0.105	0.009
σ_e^2	-0.007	0.137

Table 4

Average RMSE Across all Conditions

Parameter	Class 1	Class 2
β_1	0.42	0.43
β_2	0.23	0.25
β_4	0.20	0.20
γ	0.25	0.33
φ_{11}	0.31	0.35
φ_{22}	0.10	0.10
φ_{33}	0.07	0.08
φ_{44}	0.26	0.31
σ_e^2	0.04	0.04

Table 5

Eta-Squares from Between-Subjects ANOVA on Model Convergence Rates

	η^2 ($SS_{Effect}/SS_{CorrectedTotal}$)
S	0.033*
KL	0.413*
RV	0.487*
MD	0.005
S*KL	0.015
S*RV	0.002
S*MD	0.016
KL*RV	0.012
KL*MD	0.003
RV*MD	0.001

* $p < .05$. S: Sample size; KL: Knot Location; RV: Residual Variance; MD: Mahalanobis Distance

Table 6

Eta-Squares from Between-Subjects ANOVA for Parameter Bias

		S	RV	KL	MD	S*RV	S*KL	S*MD	RV*KL	RV*MD	KL*MD
CLASS 1	β_1	0.001	0.052	0.066	0.591*	0.000	0.021	0.066	0.027	0.054	0.005
	β_2	0.028	0.174	0.170	0.000	0.013	0.052	0.000	0.086	0.013	0.042
	β_4	0.036	0.068	0.042	0.567*	0.007	0.016	0.004	0.018	0.029	0.031
	γ	0.004	0.029	0.028	0.193	0.017	0.009	0.134	0.144	0.127	0.075
	φ_{11}	0.001	0.115*	0.106*	0.584*	0.024	0.015	0.001	0.044	0.005	0.047
	φ_{22}	0.045	0.064	0.265	0.062	0.001	0.036	0.084	0.114	0.067	0.034
	φ_{33}	0.031	0.002	0.037*	0.760*	0.029	0.010	0.024	0.020	0.006	0.057*
	φ_{44}	0.127	0.318*	0.109	0.057	0.085	0.002	0.016	0.014	0.113	0.030
	σ_e^2	0.004	0.392*	0.022	0.184*	0.000	0.039*	0.002	0.184*	0.144*	0.001
CLASS 2	β_1	0.010	0.040	0.009	0.680*	0.002	0.005	0.055	0.051	0.000	0.004
	β_2	0.024	0.023	0.191*	0.076	0.002	0.074	0.000	0.176*	0.321*	0.011
	β_4	0.001	0.022	0.000	0.585*	0.002	0.058	0.000	0.063	0.010	0.034
	γ	0.000	0.143	0.227	0.003	0.008	0.002	0.149	0.175	0.006	0.030
	φ_{11}	0.148*	0.108	0.048	0.363*	0.058	0.121	0.028	0.015	0.001	0.014
	φ_{22}	0.051	0.018	0.011	0.229	0.036	0.072	0.051	0.001	0.021	0.014
	φ_{33}	0.063	0.055	0.068	0.682*	0.000	0.002	0.001	0.046	0.007	0.000
	φ_{44}	0.288*	0.405*	0.105*	0.003	0.132**	0.001	0.001	0.036*	0.001	0.003
	σ_e^2	0.339*	0.053	0.045	0.061	0.194*	0.001	0.005	0.124	0.080	0.000

* $p < .05$. S: Sample size; KL: Knot Location; RV: Residual Variance; MD: Mahalanobois Distance

Table 7
Eta-Squares from Between-Subjects ANOVA on RMSE

		S	RV	KL	MD	S*RV	S*KL	S*MD	RV*KL	RV*MD	KL*MD
CLASS 1	β_1	0.004	0.092	0.068	0.501***	0.021	0.017	0.047	0.017	0.040	0.006
	β_2	0.042	0.039	0.410*	0.063	0.006	0.009	0.061	0.000	0.009	0.184
	β_4	0.093	0.084	0.020	0.414***	0.021	0.043	0.095	0.032	0.026	0.036
	γ	0.180**	0.494***	0.033	0.118**	0.070*	0.001	0.066*	0.004	0.000	0.004
	φ_{11}	0.002	0.382**	0.157*	0.238**	0.018	0.030	0.003	0.094*	0.010	0.000
	φ_{22}	0.042	0.147	0.429*	0.002	0.008	0.013	0.078	0.066	0.019	0.031
	φ_{33}	0.002	0.099	0.005	0.322*	0.057	0.029	0.023	0.015	0.103	0.197*
	φ_{44}	0.134	0.320*	0.079	0.043	0.020	0.007	0.024	0.001	0.206*	0.021
	σ_e^2	0.000	0.825**	0.018	0.021	0.009	0.012	0.001	0.023	0.013	0.004
CLASS 2	β_1	0.060	0.197	0.132	0.316*	0.033	0.001	0.002	0.032	0.007	0.058
	β_2	0.020	0.133*	0.440**	0.104*	0.002	0.056	0.011	0.008	0.009	0.164*
	β_4	0.032	0.030	0.007	0.303**	0.040	0.443**	0.001	0.080*	0.004	0.024
	γ	0.016	0.357*	0.264*	0.050	0.019	0.000	0.015	0.121	0.016	0.015
	φ_{11}	0.154*	0.326**	0.155*	0.066	0.048	0.071	0.055	0.016	0.016	0.007
	φ_{22}	0.133	0.238*	0.339*	0.008	0.012	0.014	0.015	0.026	0.096	0.006
	φ_{33}	0.538**	0.060	0.018	0.070	0.005	0.001	0.018	0.000	0.014	0.137
	φ_{44}	0.301**	0.377**	0.061	0.020	0.099	0.030	0.002	0.000	0.000	0.023
	σ_e^2	0.138*	0.678**	0.001	0.009	0.043	0.000	0.008	0.007	0.011	0.045

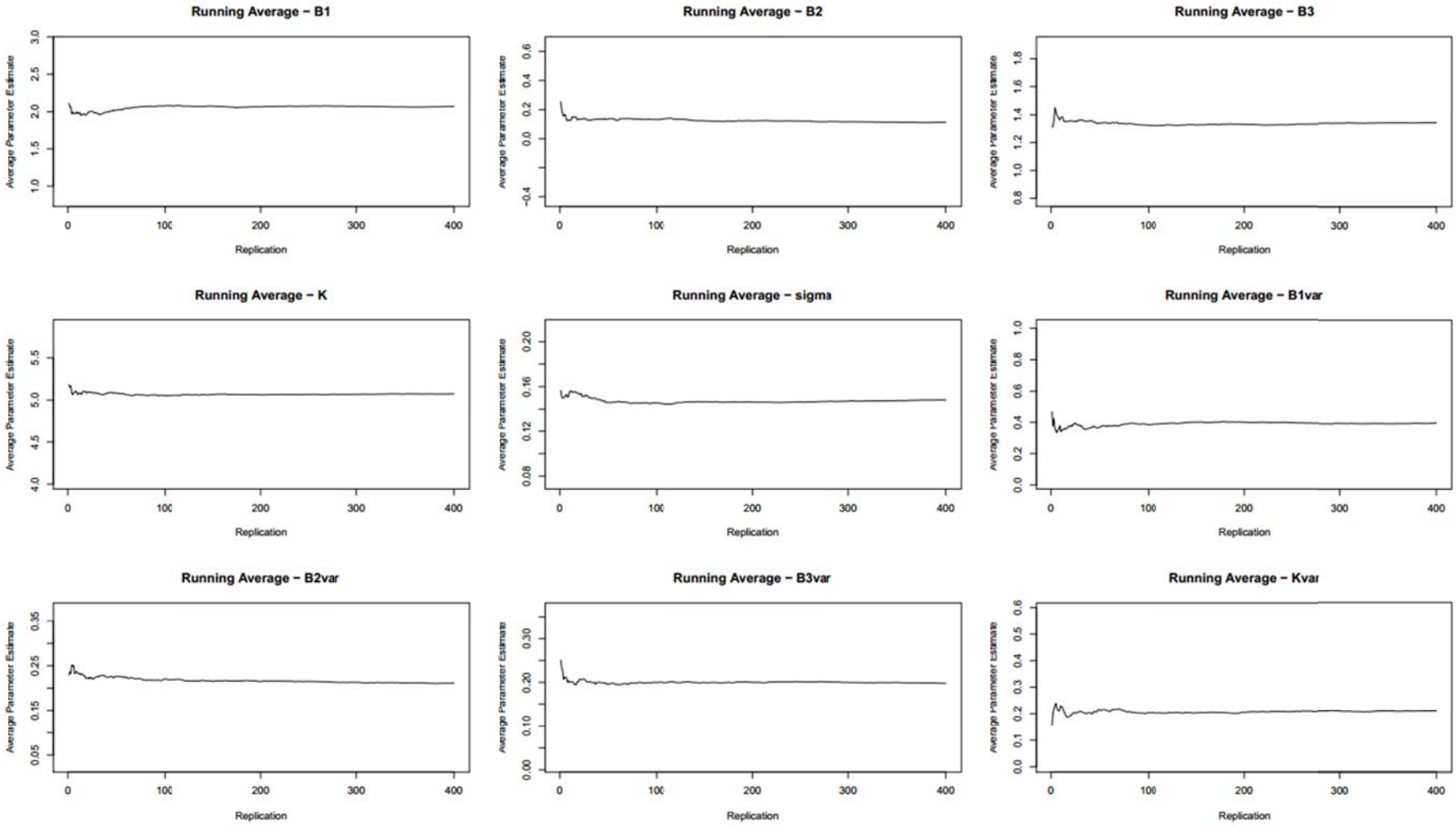
* $p < .05$. S: Sample size; KL: Knot Location; RV: Residual Variance; MD: Mahalanobis Distance

Supplemental Results from Post-hoc Simulations to Check the Sufficiency of 100 Replications per Condition

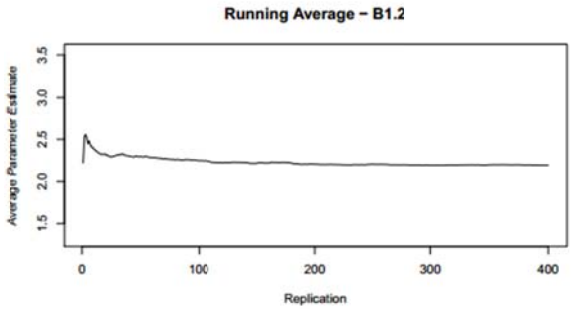
- Running Averages for Each Parameter in Class 1 and Class 2 -

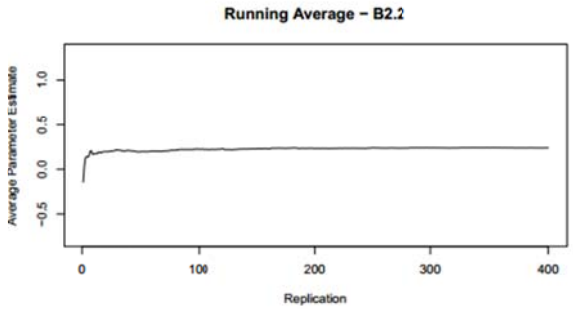
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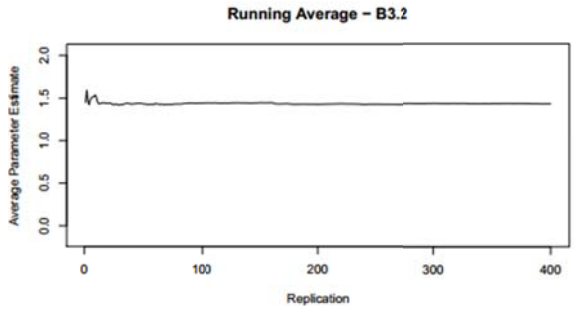
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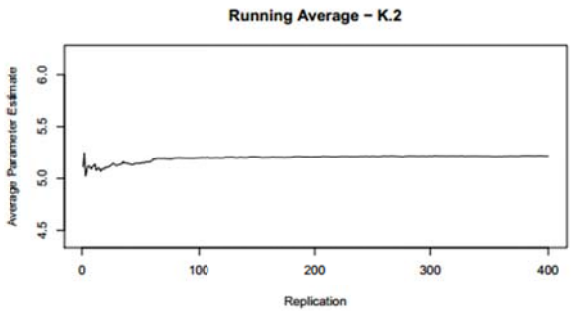


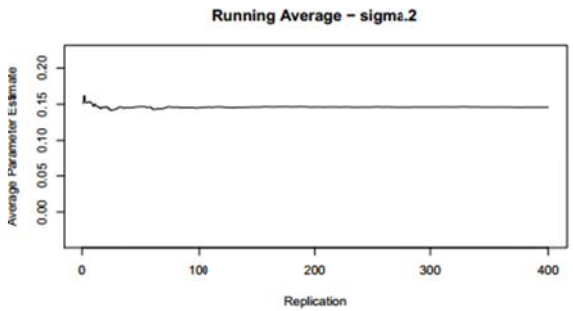
Class 2

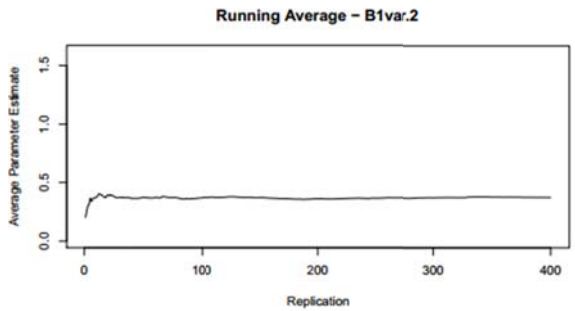

Running Average − B1.2
Average Parameter Estimate
Replication

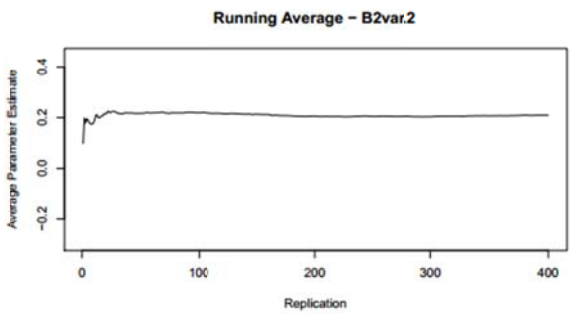

Running Average − B2.2
Average Parameter Estimate
Replication

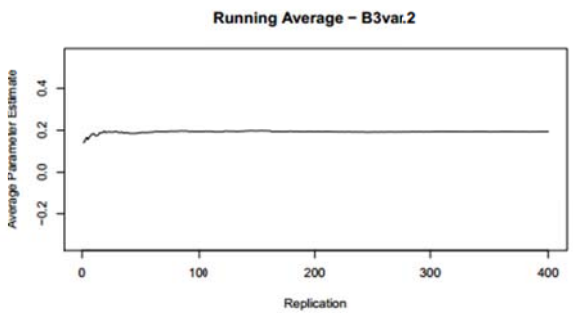

Running Average − B3.2
Average Parameter Estimate
Replication

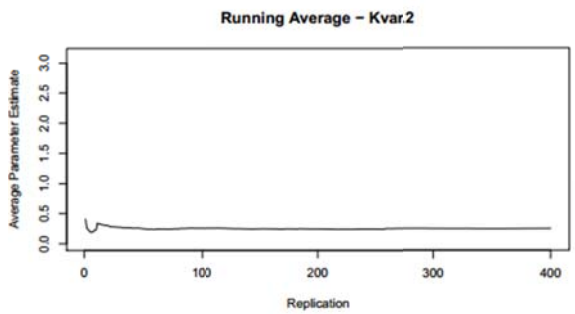

Running Average − K.2
Average Parameter Estimate
Replication


Running Average − sigma.2
Average Parameter Estimate
Replication


Running Average − B1var.2
Average Parameter Estimate
Replication

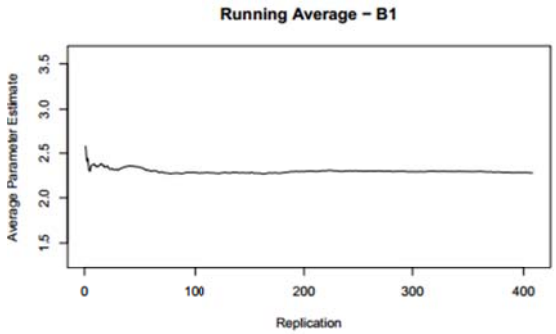

Running Average − B2var.2
Average Parameter Estimate
Replication

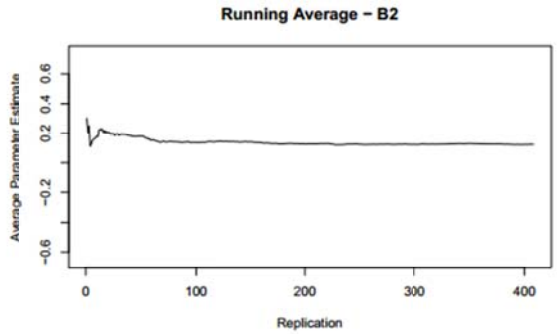

Running Average − B3var.2
Average Parameter Estimate
Replication

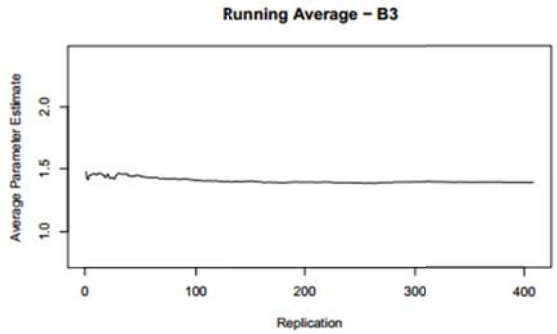

Running Average − Kvar.2
Average Parameter Estimate
Replication

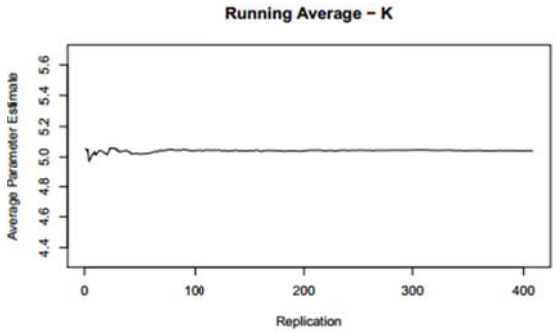
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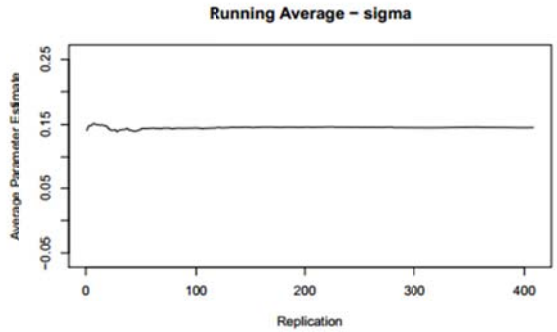
Class 1

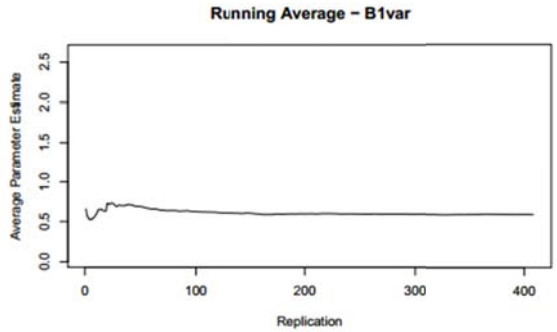


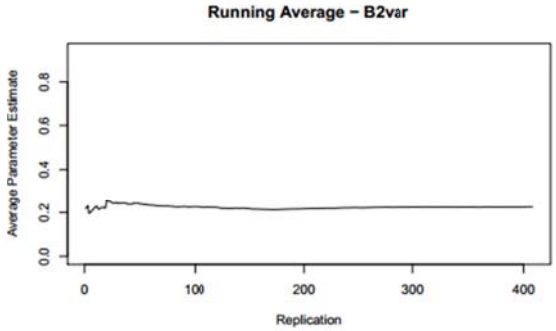


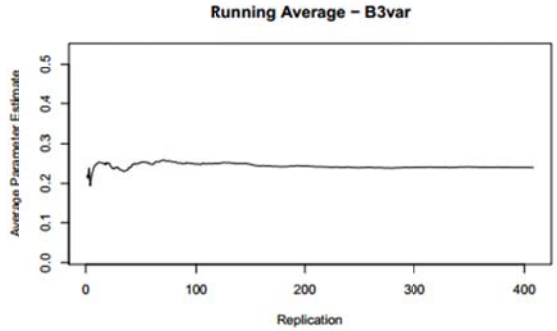


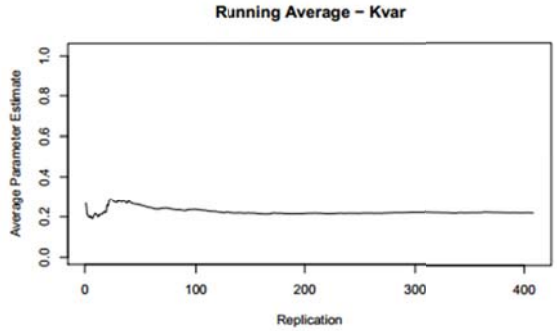




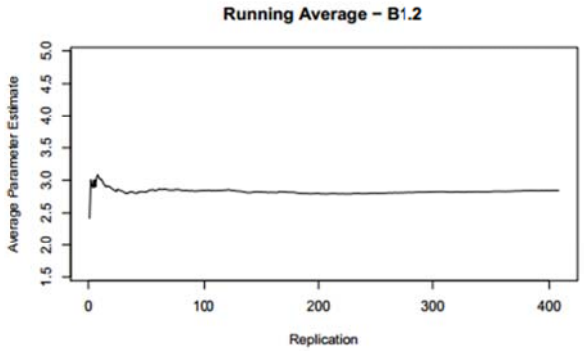


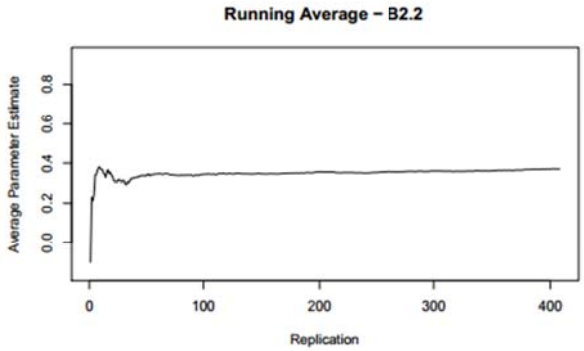


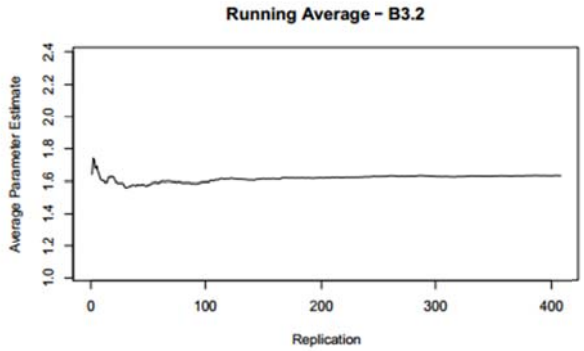


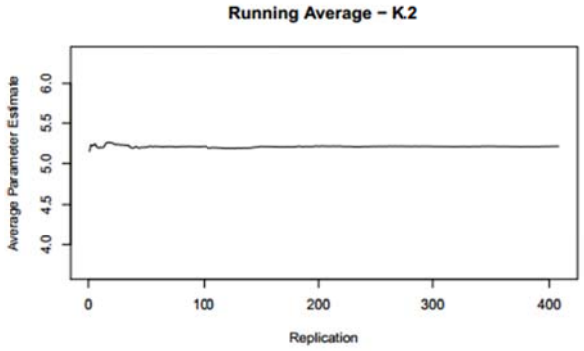


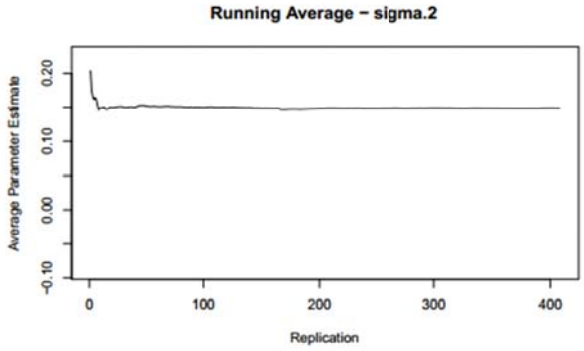
Class 2

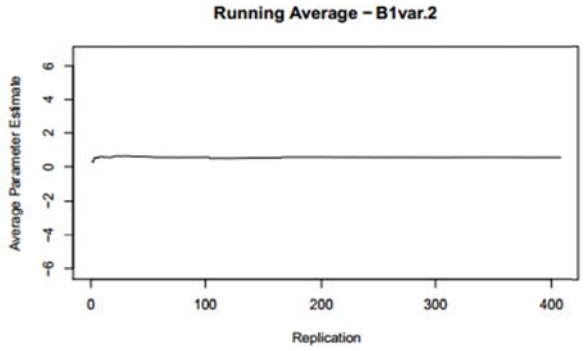

Running Average − B1.2
Average Parameter Estimate
1.5 2.0 2.5 3.0 3.5 4.0 4.5 5.0
0 100 200 300 400
Replication

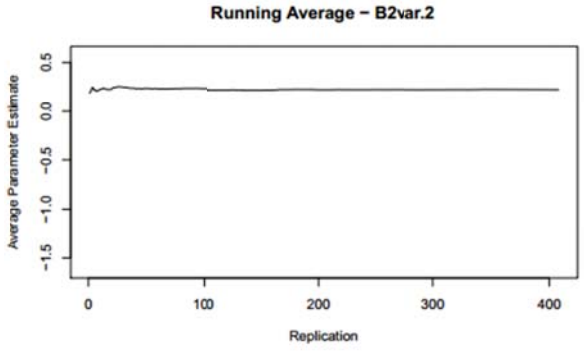

Running Average − B2.2
Average Parameter Estimate
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0 100 200 300 400
Replication

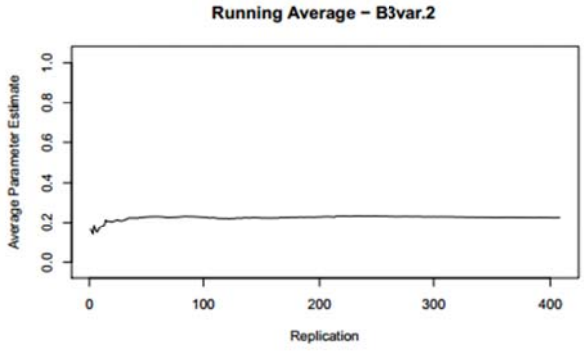

Running Average − B3.2
Average Parameter Estimate
1.0 1.2 1.4 1.6 1.8 2.0 2.2 2.4
0 100 200 300 400
Replication

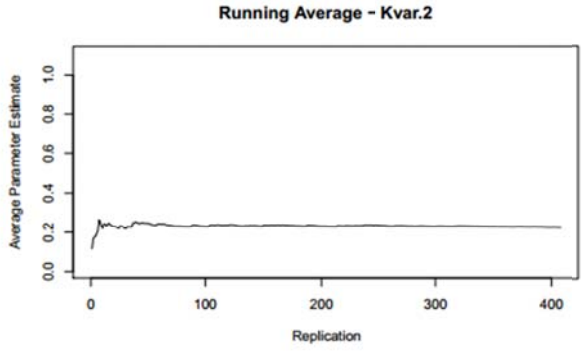

Running Average − K.2
Average Parameter Estimate
4.0 4.5 5.0 5.5 6.0
0 100 200 300 400
Replication


Running Average − sigma.2
Average Parameter Estimate
−0.10 0.00 0.10 0.20
0 100 200 300 400
Replication


Running Average − B1var.2
Average Parameter Estimate
−6 −4 −2 0 2 4 6
0 100 200 300 400
Replication


Running Average − B2var.2
Average Parameter Estimate
−1.5 −1.0 −0.5 0.0 0.5
0 100 200 300 400
Replication


Running Average − B3var.2
Average Parameter Estimate
0.0 0.2 0.4 0.6 0.8 1.0
0 100 200 300 400
Replication


Running Average − Kvar.2
Average Parameter Estimate
0.0 0.2 0.4 0.6 0.8 1.0
0 100 200 300 400
Replication